

Necati Kaan Aydınli

CONTACT

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EXPERIENCE

Software Engineering Intern | March 2026 – Present

Patientdesk AI | Y Combinator W26

- Implemented billing, subscription, and Stripe checkout workflows including reconciliation, clinic eligibility checks, and multi-clinic subscription management.
- Built and improved control plane dashboards for clinic onboarding, PMS connections, AI receptionist workflows, deployment status, observability, and platform administration.
- Optimized application performance and reliability through API caching, batch data loading, and database query improvements, reducing redundant data fetching and improving dashboard load efficiency by around 65%.

Software Engineering Intern | August 2025 – September 2025

Artifeye Technologies

- Developed a local Retrieval Augmented Generation (RAG) agent for processing and analyzing large scale textile manufacturing documents.
- Built ChromaDB-based vector retrieval pipelines and custom indexing/query workflows, improving query efficiency by around 30% and achieving 1000 token retrieval in under 100ms.

EDUCATION

Bilkent University

Bachelor of Science in Computer Engineering

- GPA: 3.38
- President, Association for Computing Machinery (ACM)
 - Lead a 12 member team and organized 10+ technical events for a community of 1,000+ students.
- Bilkent Campus Fellow at fal.ai

PROJECTS

LogicDrawer - AI Powered Digital Logic Design Platform | www.logicdrawer.app

TypeScript, Node.js/Express, MongoDB, Python, YOLO

- Developed and deployed a web application for designing and simulating digital logic circuits, currently serving 700+ users, featuring a drag and drop interface and comprehensive component library.
- Trained a custom YOLO model achieving ~90% mAP@0.5 on a 7K image dataset for detecting handdrawn logic circuits.
- Engineered an integrated AI agent ecosystem with diverse tool support, enabling users to generate complex logic circuits entirely through natural language commands.
- Implemented multibit simulation and logic evaluation, supporting automatic truth table and Karnaugh map generation from circuits.
- Used OpenCV library to detect arrangements of components with their connections.
- Designed and implemented a Verilog parser and compiler to transform HDL code into interactive visual circuit representations for simulation, validation, and verification workflows.

BilRev - Automated Paper Review System

TypeScript, Node.js, Playwright, Supabase, FastAPI

- Built a web platform that connects to Overleaf and runs AI driven paper reviews with inline comments.
- Conducted requirements analysis and system design for a modular automation system.
- Developed a modular Playwright automation layer to integrate with Overleaf without using public APIs.

ChromaDB Node for n8n - Fully local vector database

TypeScript, n8n, ChromaDB, LangChain

- Developed a custom n8n node to integrate ChromaDB for storage, retrieval, and similarity search.
- Implemented client configuration for both self hosted and cloud deployments

SKILLS

Programming Languages: TypeScript, Java, Python, JavaScript, C++, HTML, CSS

Frameworks & Libraries: JavaFX, Express.js, Vite, Electron, Next.js, React

AI & Systems: RAG, Vector Databases, YOLO object recognition

Tools & Technologies: Supabase, MongoDB, Git, ChromaDB, Ollama, Docker, Playwright

LANGUAGES & INTERESTS

Languages: Turkish (native), English (fluent)

Interests: Water Polo, Chess, Kickboxing, Swimming, Badminton